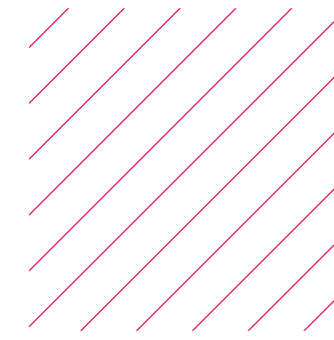




CI INSTITUTE OF NURSING

CERTIFIED NURSE
ASSISTANT TRAINING PROGRAM



Module 13: Long Term Patient/Resident

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MODULE 13: LONG TERM PATIENT/RESIDENT

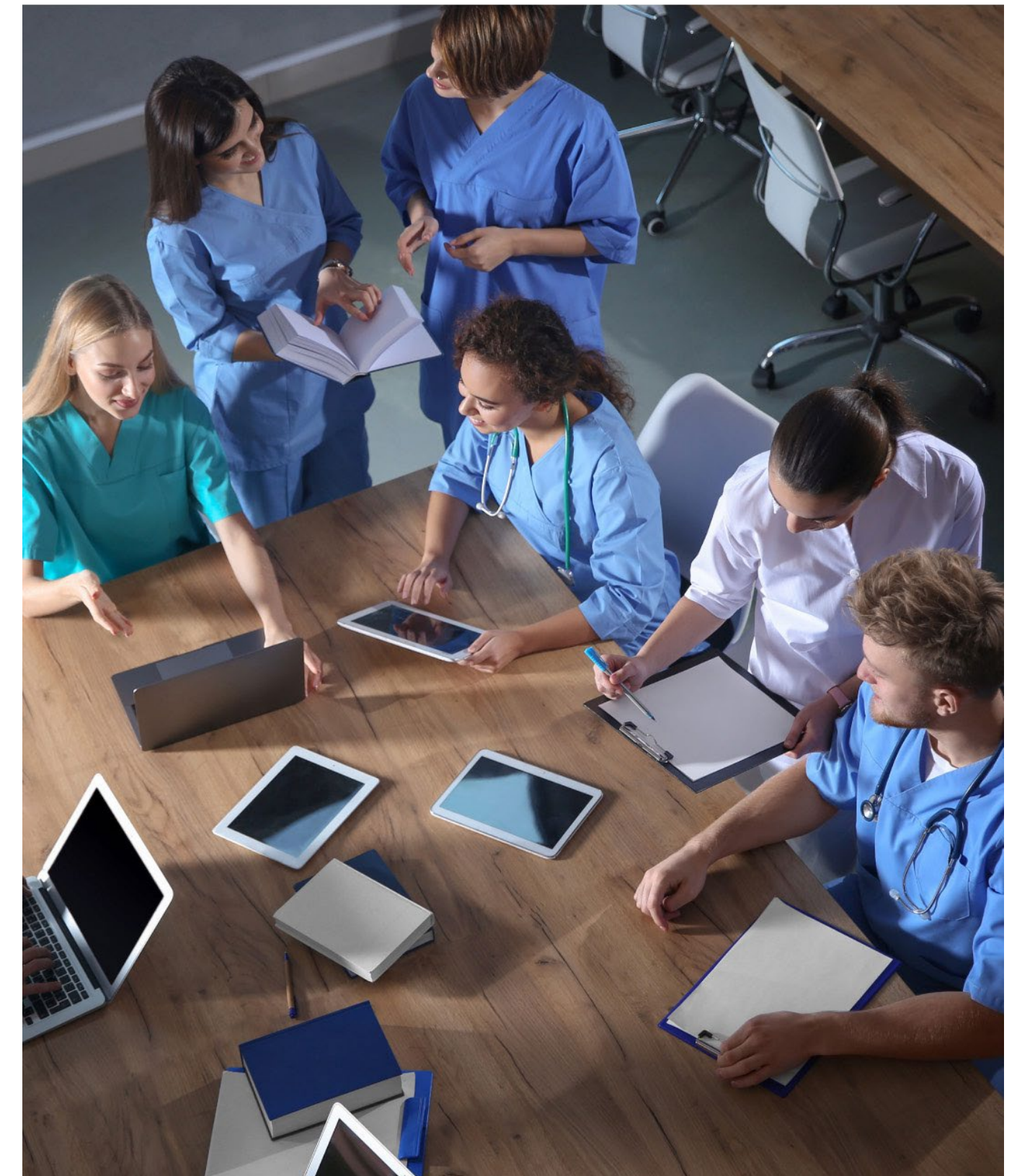
Minimum Number of Theory Hours:

13

Recommended Clinical Hours:

4

Module 13: Long Term Patient/Resident

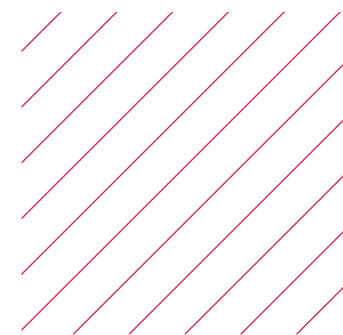


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STATEMENT OF PURPOSE

The purpose of this unit is to introduce the student to the basic structure of the body and to review the effect of aging on body structure and function. Common physical and psychological conditions found in elderly patients are presented along with approaches to care. Community resources commonly available to assist elderly patients with their psychological, recreational, and social needs are presented.



TERMINOLOGY:

Long Term Care Resident:

1. Acute
2. Assisted living
3. Chronic
4. Debilitating
5. Dementia
6. Long-term care
7. Reality orientation
8. Reminiscing
9. Patients/residents
10. Skilled nursing facility
11. Subacute care
12. Validation therapy

Medical and Anatomical Terms:

13. Anatomic position
14. Anatomy
15. Anterior
16. Cells
17. Connective tissue
18. Contractures
19. Disease
20. Distal

21. Epithelial tissue

22. Health

23. Inferior

24. Joints

25. Lateral

26. Membranes

27. Muscle tissue

28. Nerve tissue

29. Organ

30. Peritoneum

31. Physiology

32. Posterior

33. Prefix

34. Proximal

35. Suffix

36. Superior

37. System

38. Traction

Nervous System:

39. Age-related Macular Degeneration (AMD)

40. Agitated

41. Alzheimer's disease

42. Aphasia

43. Autonomic dysreflexia

44. Bipolar disorder

45. Brain

46. Cataract

47. Catastrophic reaction

48. Cerebral palsy

49. Cerebral Vascular Accident (CVA)

50. Concussion

51. Confusion

52. Delirium

53. Delusion

54. Dementia

55. Developmentally disabled

56. Disruptive behavior

57. Elopement

58. Epilepsy

59. Generalized anxiety

60. Glaucoma

61. Hallucination

62. Hemianopsia

63. Hemiplegia



Patient, resident, and client are synonymous terms referring to the person receiving care

TERMINOLOGY:

- 64. Multiple sclerosis
- 65. Nerve
- 66. Neuron
- 67. Otosclerosis
- 68. Panic disorder
- 69. Paralysis
- 70. Paranoia
- 71. Paranoid schizophrenia
- 72. Paraplegia
- 73. Parkinson's disease
- 74. Peripheral nervous system
- 75. Post-traumatic stress syndrome
- 76. Quadriplegia
- 77. Reality orientation
- 78. Seizure
- 79. Stroke
- 80. Sundowner syndrome
- 81. Transient Ischemic Attack (TIA)
- 82. Tremor
- 83. Vertigo

Immune System:

- 84. Antibody
- 85. Antigen
- 86. Auto Immunodeficiency Syndrome (AIDS)
- 87. Carrier
- 88. Human Immunodeficiency Virus
- 89. Immunity
- 90. Lymph node

- 91. T-cell
- 92. White blood cell

Cardiovascular System:

- 93. Anemia
- 94. Angina
- 95. Anti-embolic stockings
- 96. Artery
- 97. Atherosclerosis
- 98. Blood
- 99. Capillary
- 100. Congestive Heart Failure (CHF)
- 101. Coronary artery disease
- 102. Dyspnea
- 103. Edema
- 104. Embolism
- 105. Heart
- 106. Hypertension
- 107. Hypoxia
- 108. Ischemia
- 109. Myocardial infarction
- 110. Orthopnea
- 111. Pacemaker
- 112. Peripheral vascular disease (PVD)
- 113. Phlebitis
- 114. Pulmonary edema
- 115. Sequential compression device (SCD)
- 116. Systole
- 117. Thrombus
- 118. Varicose vein

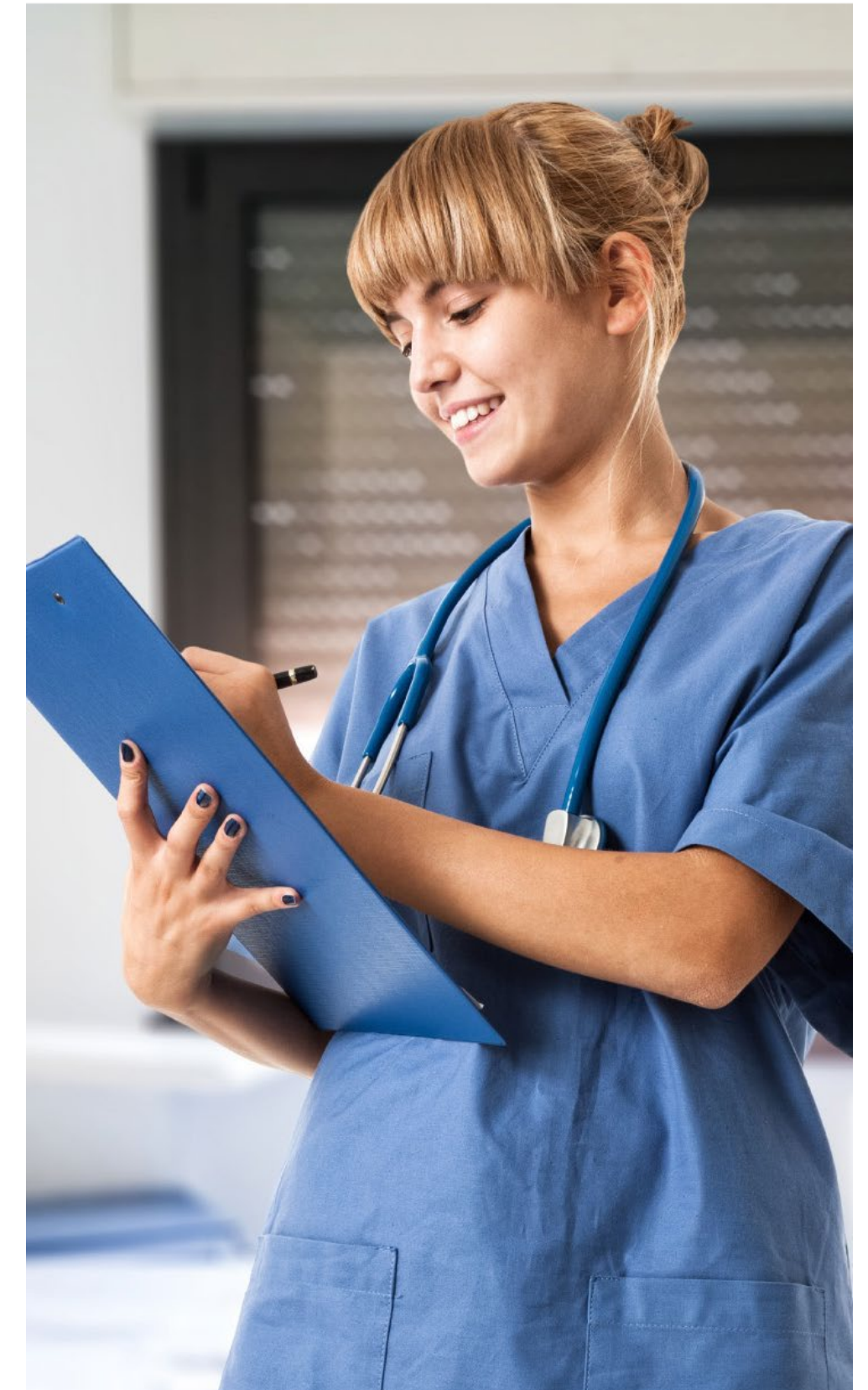
- 119. Vein
- 120. Vessel

Integumentary System:

- 121. Abrasion
- 122. Allergies
- 123. Bony prominences
- 124. Cyanotic
- 125. Debride
- 126. Decubitus ulcer
- 127. Dermis
- 128. Epidermis
- 129. Excoriation
- 130. Friction
- 131. Gangrene
- 132. Hematoma
- 133. Lesion
- 134. Necrosis
- 135. Pallor
- 136. Pediculosis
- 137. Pressure ulcer
- 138. Pruritus
- 139. Psoriasis
- 140. Scabies
- 141. Shearing
- 142. Skin
- 143. Skin tears

Gastrointestinal System:

- 144. Absorption
- 145. Bile
- 146. Bowel movement
- 147. Colon
- 148. Colostomy



TERMINOLOGY:

- 149. Constipation
- 150. Defecation
- 151. Diarrhea
- 152. Digestion
- 153. Diverticulitis
- 154. Elimination
- 155. Enema
- 156. Fecal impaction
- 157. Feces
- 158. Flatus
- 159. Gastrectomy
- 160. Gastric tube
- 161. Hernia
- 162. Ileostomy
- 163. Nasogastric tube (NG)
- 164. Obstruction
- 165. Occult blood
- 166. Ostomy
- 167. Percutaneous Endoscopic Gastrostomy (PEG)
- 168. Peristalsis
- 169. Stoma
- 170. Stool
- 171. Suppository
- 172. Ulcer
- 173. Urgency
- 174. Urostomy

Respiratory System:

- 175. Alveoli
- 176. Asthma
- 177. Bronchi
- 178. Carbon dioxide

- 179. Chronic Heart Failure (CHF)
- 180. Chronic Obstructive Pulmonary Disease (COPD)
- 181. Dyspnea
- 182. Emphysema
- 183. Expiration
- 184. Inspiration
- 185. Larynx
- 186. Lungs
- 187. Mucous
- 188. Nasal cannula
- 189. Nebulizer
- 190. Orthopnea
- 191. Oxygen mask
- 192. Pharynx
- 193. Pneumonia
- 194. Sputum
- 195. Tracheostomy
- 196. Tuberculosis (TB)
- 197. Upper respiratory infection (URI)

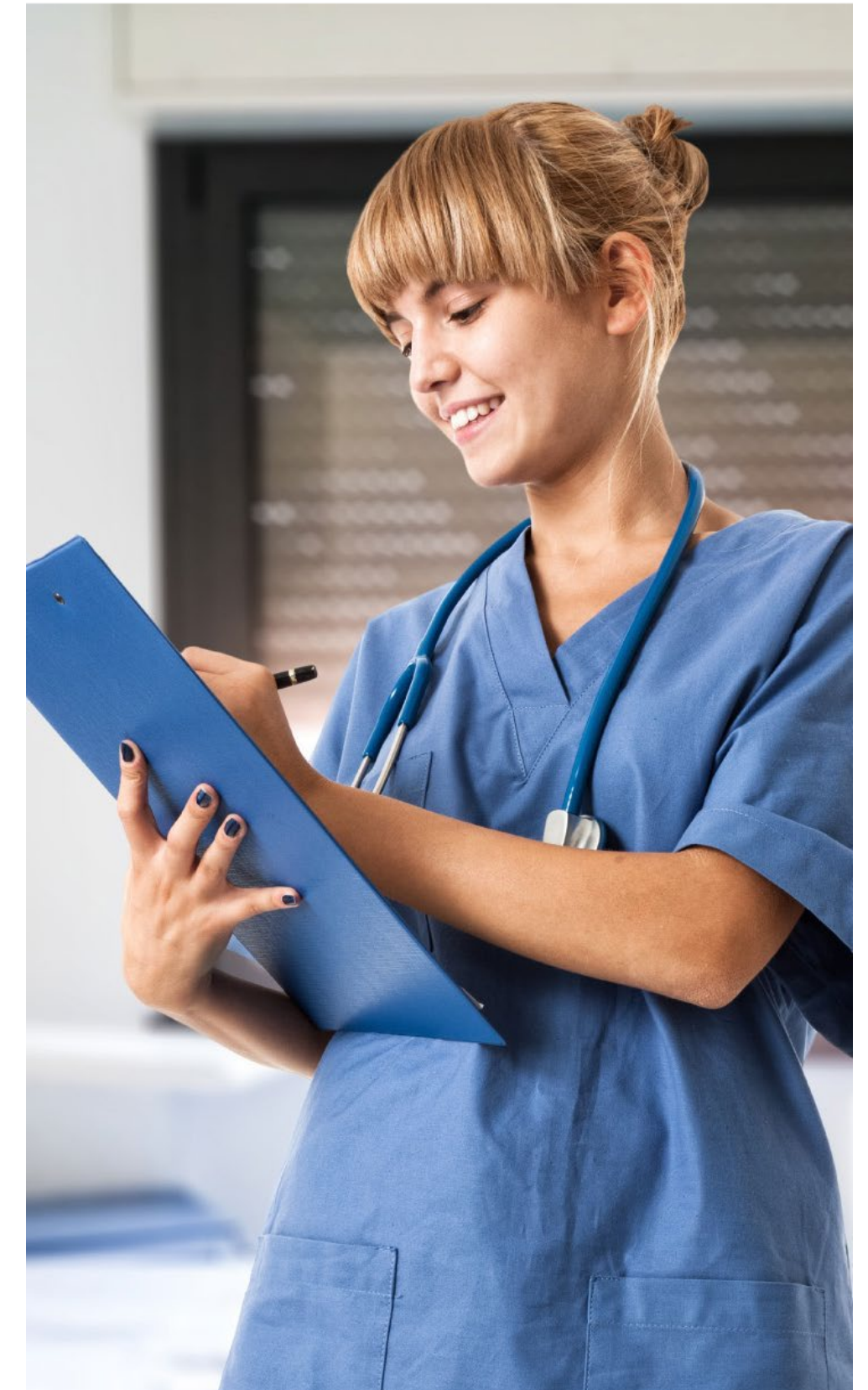
Musculoskeletal System:

- 198. Abduction
- 199. Adduction
- 200. Amputation
- 201. Arthritis
- 202. Atrophy
- 203. Bursitis
- 204. Contracture
- 205. Extension
- 206. Flexion
- 207. Fracture

- 208. Full weight-bearing (FWB)
- 209. Gout
- 210. Muscle
- 211. Muscular dystrophy
- 212. Osteoarthritis
- 213. Osteoporosis
- 214. Phantom pain
- 215. Pronation
- 216. Prosthesis
- 217. Range of Motion (ROM)
- 218. Rheumatoid arthritis
- 219. Rotation
- 220. Skeleton
- 221. Supination
- 222. Total hip arthroplasty
- 223. Trapeze
- 224. Vertebrae

Urinary System

- 225. Catheter
- 226. Calculi
- 227. Chronic renal failure
- 228. Clean catch
- 229. Condom catheter
- 230. Cystitis
- 231. Dialysis
- 232. Diuresis
- 233. Dysuria
- 234. End-Stage Renal Disease (ESRD)
- 235. Hematuria
- 236. Indwelling catheter
- 237. Kidney



TERMINOLOGY:

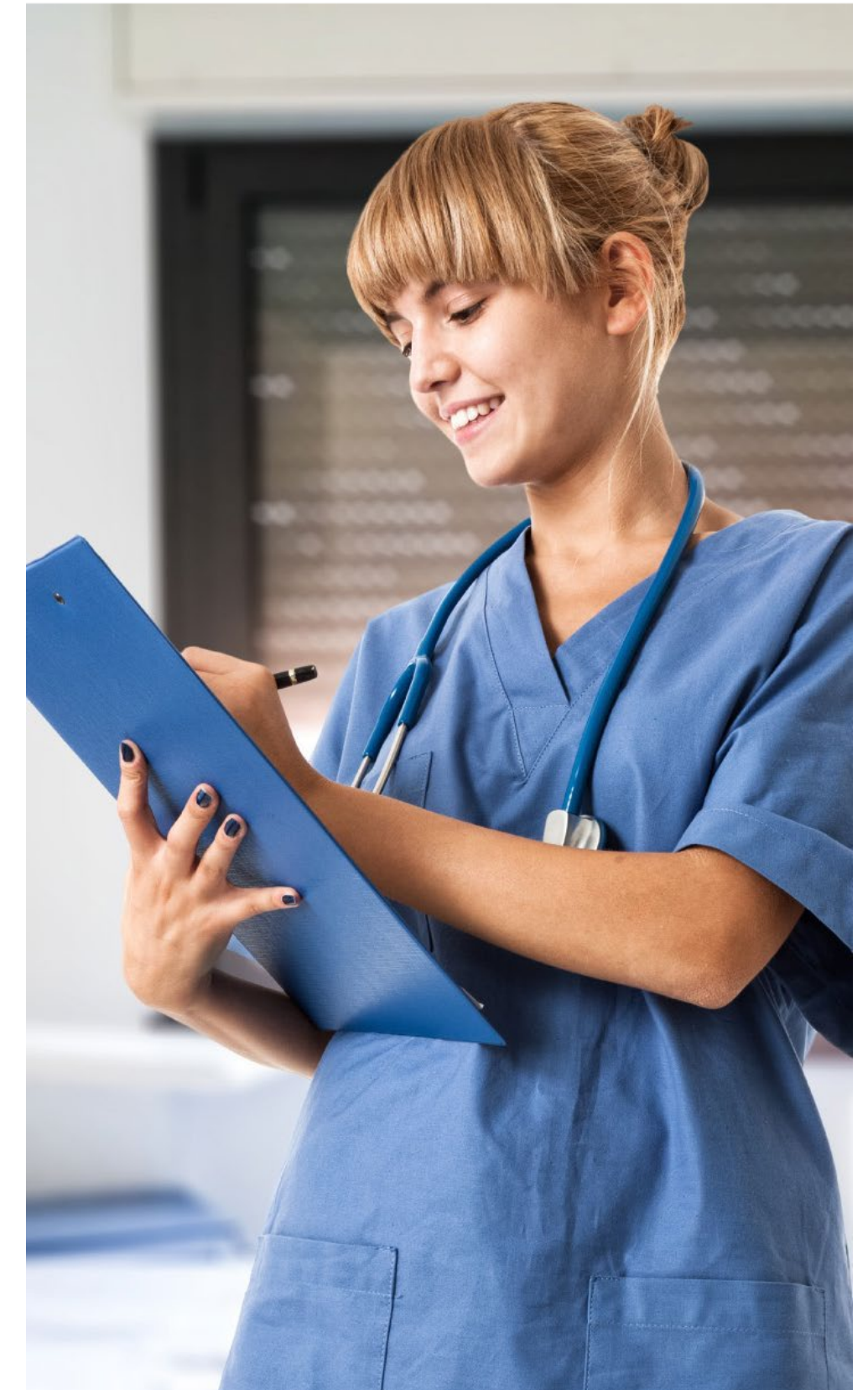
- 238. Micturition
- 239. Nephritis
- 240. Renal calculi
- 241. Retention
- 242. Specimens
- 243. Ureter
- 244. Urethra
- 245. Urinalysis
- 246. Urinary bladder
- 247. Urinary drainage bag
- 248. Urinary incontinence
- 249. Urinary Tract Infection (UTI)
- 250. Void

Reproductive System:

- 251. Benign prostatic hypertrophy
- 252. Chlamydia
- 253. Cystocele
- 254. Douche
- 255. Genitalia
- 256. Genital herpes

- 257. Human Papilloma Virus(HPV)
- 258. Gonorrhea
- 259. Hemorrhoid
- 260. Herpes simplex 2
- 261. Hormones
- 262. Hysterectomy
- 263. Mammogram
- 264. Mastectomy
- 265. Pelvis Inflammatory disease (PID)
- 266. Penis
- 267. Prostate gland
- 268. Rectocele
- 269. Sexually transmitted disease (STD)
- 270. Syphilis
- 271. Testes
- 272. Trichomoniasis
- 273. Uterus
- 274. Vagina

- 275. Venereal
 - 276. Vulvovaginitis
- ## **Endocrine System:**
- 277. Diabetes mellitus
 - 278. Fasting Blood Sugar (FBS)
 - 279. Glucose
 - 280. Glycosuria
 - 281. Goiter
 - 282. Hormones
 - 283. Hyperglycemia
 - 284. Hypoglycemia
 - 285. Insulin
 - 286. Insulin-Dependent Diabetes Mellitus (IDDM)
 - 287. Metabolism
 - 288. Non-Insulin-Dependent Diabetes Mellitus (NIDDM)
 - 289. Pituitary gland
 - 290. Thyroxine



Patient, resident, and client are synonymous terms referring to the person receiving care



PERFORMANCE STANDARDS

Upon completion of five (5) hours SNF/ICF* or three (3) non-SNF/ICF of class plus homework assignments and four (4) hours of clinical experience, the learner will be able to:**

1. Define key terminology
2. Describe common basic human needs, changes, and interventions for the elderly patient/resident; environmental/physical, psychological, social/behavioral, recreational and spiritual
3. Describe common community resources to meet the needs of the elderly
4. Describe the special needs and changes of persons with developmental and mental disorders including intellectual disability, cerebral palsy, epilepsy, Parkinson's disease, and mental illness
5. Describe the special needs of persons with Alzheimer's Disease and other related dementias
6. Describe the body's basic organization and composition
7. List the body systems, including basic anatomy and physiology, common diseases of the elderly with signs and symptoms, Nurse Assistant duties and observations, aging changes and complications of immobility
8. Describe changes in body systems associated with aging



/// DEFINE KEY TERMINOLOGY

- A. Review the terms listed in the terminology section
- B. Spell the listed terms accurately
- C. Pronounce the terms correctly
- D. Use the terms in their proper context



Describe common basic human needs, changes, and interventions for the elderly patient/resident; environmental/physical, psychological, social/behavioral, recreational and spiritual

A. Environmental/physical

1. Safety

- a. More light for better vision
- b. Freedom from hazards
- c. Support (adaptive devices)
- d. Good personal hygiene

2. Shelter

- a. Appropriate living situation
- b. Environmental control (heating, air conditioning)

3. Nourishment

- a. Balanced nutrition
- b. Adequate fluids

Describe common basic human needs, changes, and interventions for the elderly patient/resident; environmental/physical, psychological, social/behavioral, recreational and spiritual

B. Psychological

1. Maintain self-esteem
 - b. Respectful treatment
 - c. Recognition of individual differences
2. Adjustment to role change and loss of independence
 - b. Respect privacy
 - c. Give choices
 - d. Encourage activities

Describe common basic human needs, changes, and interventions for the elderly patient/resident; environmental/physical, psychological, social/behavioral, recreational and spiritual

C. Social

1. Social interactions

- b. Encourage family involvement
- c. Encourage social/community activities

2. Adjustment to losses

- b. Death of spouse/friends
- c. Encourage verbalization, such as reminiscing
- d. Encourage new social contacts

3. Financial

- b. Changes in income interfere with meeting basic needs of security, love, belonging and self-esteem
- c. Refer to social service

Describe common basic human needs, changes, and interventions for the elderly patient/resident; environmental/physical, psychological, social/behavioral, recreational and spiritual

D. Recreational

1. Diversion
 - b. Encourage hobbies
 - c. Involve in facility/community events
 - d. Encourage family involvement
2. Self-esteem
 - a. Encourage involvement
 - b. Give compliments
 - c. Reinforce positive traits or abilities

E. Spiritual

1. Provide respect for individual choices
2. Provide opportunities to participate in religious activities

Describe common community resources to meet the needs of the elderly.

- A. Area Agency on Aging-AAA, (Request Community Resource Manual)**
- B. Adult Day Care Center**
- C. Support groups**
 - 1. American Diabetes Association
 - 2. Braille Institute or Blind Center
 - 3. Alzheimer's Society
 - 4. Parkinson's Support
 - 5. Multiple Sclerosis Support
 - 6. Muscular Dystrophy
 - 7. Hospice/Bereavement
- D. County Health Center**
- E. Community Hospitals and their hospice**
- F. Internal Revenue Service (IRS) information**
- G. Long Term Care Ombudsman**
- H. Meals onWheels**
- I. Medicare Office (see Social Security Office)**
- J. Mental Health Department**
- K. National Alliance for the Mentally Ill (NAMI)**
- L. American Disability Act**
- M. Adult Protective Services (APS)**
- N. Red Cross/Blood Pressure Clinic**
- O. Senior Center**
- P. Senior Housing**
- Q. Social Security Office Suicide prevention**
- R. Home health agencies**
- S. Voter registration**
- T. American Association of Retired Persons (AARP)**
- U. Fraternal and social organizations**
- V. Regional centers**
- W. Churches/religious organizations**

Describe the body's basic organization and composition.

A. Anatomy-Structure; Physiology-Function

B. The Cell

1. The basic unit of body structure; building blocks of the body
2. Function, size, and shape of cells may differ
3. Cells need food, water, and oxygen to survive

C. Tissues

1. Groups of cells with similar functions. Four basic types of tissues in body
2. Epithelial tissue – protective
3. Connective tissue – support and connect
4. Muscle tissue – shorten and lengthen
5. Nerve tissue – carry electrical impulses

D. Organs

1. Made up of different types of tissues
2. Perform special functions

E. Systems

1. Groups of organs that work together to perform specific functions
2. Ten major body systems
 - a. Integumentary System
 - b. Respiratory System
 - c. Cardiovascular System
 - d. Musculoskeletal System
 - e. Endocrine System
 - f. Nervous System
 - g. Gastrointestinal System
 - h. Urinary System
 - i. Reproductive System
 - j. Immune System

F. Health; state of mental, physical and social well-being

G. Disease; any change from the healthy state



List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

A. Integumentary system (skin)

1. Anatomy

a. Epidermis is the top surface layer

- 1) It is primarily dry, dead cells that shed continuously
- 2) Thin layer with no blood supply
- 3) Functions as a barrier to water loss and prevention infection

b. Dermis is the second layer

- 1) Varies in thickness in different areas of the body
- 2) Naturally very moist
- 3) Gives skin its strength
- 4) Contains the blood vessels, lymph vessels, nerve endings, oil glands, and hair follicles
- 5) Function of dermis is to aid in body temperature control and is the origin of our sense of touch, pressure, and pain

c. Subcutaneous tissues are deeper tissues beneath the dermis

- 1) Tissues are fat and connective tissue
- 2) These tissues do not tolerate lack of oxygen and cell death results easily



List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

2. Functions

- a. Provides protective barrier against microorganisms and infection
- b. Provides sensation through nerve endings
- c. Shields body tissue from injury
- d. Regulates body temperature
- e. Eliminates waste products
- f. Produces Vitamin D for body use
- g. Prevents loss of too much water
- h. It is the largest “organ” of the body

3. Diseases and Disorders

- a. Skin Lesions/Wounds - change in skin structure caused by injury, trauma, aging, or disease
 - 1) Signs and Symptoms
 - a) Rash
 - b) Raised spots filled with pus or fluid
 - c) Irregular reddened areas that itch (Pruritus)
 - d) Dry crusts; scabs



List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

- e) Break in skin integrity
- 2) Nurse Assistant duties and observations
 - a) Observation and reporting of any skin abnormalities
 - b) Observe drainage on dressing
 - c) Soap and water are contraindicated
 - d) Tepid water when bathing
 - e) Gloves
 - f) Do not remove crusts
 - g) Notify licensed nurse if:
 - Skin lesions drain
 - Nature of drainage changes, increases
 - Drainage has odor
 - Drainage changes color
 - Dressing needs changing
 - Wound has signs of redness, red streaks, heat, pus, drainage

List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

b. Pressure Sores (Decubiti) - break in skin integrity that develops over bony prominence as result of pressure

1) Signs and Symptoms a) Stage One

- Redness over bony prominence (dark skinned people may appear blue or black) that does not go away after 30 minutes
- Will reverse if precautions are taken

b) Stage Two

- Reddened skin with abrasions, blisters or shallow crater
- Epidermis and/or dermis may both be involved

c) Stage Three

- All layers of skin destroyed
- Deep crater forms

d) Stage Four

- Ulcer extends through skin and subcutaneous tissue
- Bone, muscle, and underlying structures may be involved

List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

2) Nurse Assistant duties and observations

- a) Change position at least every two hours
- b) When sitting, teach patient/resident to shift every ten (10) minutes
- c) Proper nutrition and hydration
- d) Immediate removal of feces or urine
- e) Inspect skin area whenever personal care is given, especially over bony prominences
- f) Remove wrinkles and foreign objects from bed linens
- g) Avoid hot water and friction
- h) Lubricate skin with lotions
- i) Separate body areas to avoid rubbing
- j) Mechanical aids
 - Sheepskin
 - Alternating-pressure mattress
 - Pillows for bridging
 - Gel and air cushions
 - Heel protectors



List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

- k) Use turning sheet to move dependent patients/residents in bed
- l) Elevate HOB no more than 30 degrees
- m) Check improperly fitted braces and restraints
- n) Check tubing for rubbing
- o) Know signs and symptoms of pressure sore, and report immediately to licensed nurse
- p) PREVENT: much easier to prevent than treat
- c. Pediculosis, Lice and Scabies: tiny parasites that are difficult or impossible to see
 - 1) Signs and Symptoms
 - a) Intense itching
 - Pediculosis: in areas of hair growth
 - Scabies: follows blood vessels, commonly seen in webs of fingers, inside wrists and elbows, underarms and waist and nipple areas
 - Rash
 - 2) Nurse Assistant duties and observations
 - a) Avoid direct contact; gloves
 - b) Report any signs or symptoms of disorder



List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

- c) Thorough washing of linen
- d) Medicated shampoos/lotions may be used
- d. Other related subjects (optional); Kaposi's Sarcoma, Monkey Pox, burns, impetigo

B. Respiratory System

1. Anatomy & Physiology

- a. Nose and mouth
- b. Trachea
- c. Bronchi
- d. Lungs
- e. Alveoli

2. Functions

- a. Lifeline of the body
- b. Bring oxygen into the body and eliminate carbon dioxide from the body

List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

3. Diseases and Disorders

a. Upper respiratory infection (URI): infection in nose, sinuses, and throat

1) Signs and Symptoms

- a) Fever
- b) Runny nose and eyes
- c) Cough
- d) Sore throat

2) Nurse Assistant duties and observations

- a) Encourage rest
- b) Encourage fluids
- c) Dispose of tissues promptly to avoid spread
- d) Report to licensed nurse:
 - Fever
 - Dyspnea

List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

- Change in rate or rhythm of respiration
 - Change in mucus color from clear to green, yellow or blood-tinged
 - Pallor, cyanosis
 - Signs that infection is moving to chest or lungs
- b. Pneumonia: infection/inflammation of the lungs often seen in patients/residents with compromised immune system
- 1) Signs and symptoms
 - a) Same as URI, only more severe
 - b) Chest pressure and discomfort
 - c) Extreme fatigue/weakness
 - d) Cough
 - 2) Nurse Assistant duties and observations
 - a) As above for URI
 - b) Monitor vital signs carefully, especially temperature & respiration
 - c) Elevate head of bed/orthopneic position, as needed
- c. COPD (Chronic Obstructive Pulmonary Disease) - a condition that results in chronic obstruction of the respiratory system that is not reversible



List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

1) Signs and symptoms

a) Asthma: breathing disorder caused by reaction to allergen

- Wheezing
- Tightened airways
- Mucus blocks passageways
- Labored breathing
- Frequent coughing

b) Chronic Bronchitis: prolonged inflammation in the bronchi due to infection or irritants • Persistent cough

- Sputum production (especially in the morning)
- Respiratory distress

c) Emphysema: occurs when air flow is obstructed at the level of the alveoli

- Can bring O₂ into lungs, but can't expel CO₂
- Triggered by pollutants, smoke and cold
- Frequent lung infections
- Extreme dyspnea
- Loss of appetite/weight loss

List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

- Irritability
- Air hunger

2) Nurse Assistant duties and observations

- a) Teach pursed-lipped breathing (to increase expiration of CO₂)
- b) Calm patient/resident when anxious (to decrease O₂ need)
- c) Raise head of bed/orthopneic position
- d) Oxygen therapy (O₂ precautions)-Do not change Low O₂ settings)
- e) Nutrient dense foods; frequent snacks, not meals
- f) Encourage fluids
- g) Pace activities
- h) Avoid raising arms over head
- d. Tuberculosis

1) Definition

- a) A highly contagious lung disease caused by a bacterium that is carried on mucous droplets suspended in the air
- b) Two types of TB

List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

- Latent TB - Someone with latent TB shows no symptoms
- Active TB – Shows symptoms. Occurs when tuberculosis enters the body and the inactive tubercle breaks down; bacteria multiply and spread to other parts of the body

2) Signs and symptoms

- a) Fatigue
- b) Night sweats
- c) Coughing up blood (hemoptysis)
- d) Prolonged coughing
- e) Slight fever
- f) Chills
- g) Shortness of breath and/or dyspnea
- h) Loss of appetite and weight

3) Nurse Assistant duties and observations

- a) Annual PPD (Mantoux test)
- b) Chest x-ray every two years if positive skin test
- c) Use of barriers if productive cough

List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

d) Discard tissues, sputum promptly

4. Other related subjects (optional): tracheostomy, esophageal speech and electronic speech, SARS, malignancies

C. Cardiovascular System

1. Anatomy and Physiology

a. Heart; pumps blood which carries oxygen and nutrients to cells

b. Blood

1) Transports oxygen, food and hormones to cells

2) Removes carbon dioxide and other waste products from cells

3) Controls pH level and body temperature

4) Clots the blood and fights pathogens and poisons

c. Blood vessels

1) Arteries carry blood away from the heart

2) Veins carry blood towards the heart

3) Capillaries; smallest vessels that connect arteries and veins

List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

2. Functions

- a. Transportation system that delivers nutrition and oxygen to the cells and takes away waste products
- b. Closed system is kept in motion by the force of the heart

3. Common disease and disorders

- a. Coronary Artery Disease (CAD) occurs when the coronary arteries narrow, causing reduced blood supply to the heart depriving the heart muscle with oxygen and nutrients

1) Signs and Symptoms

- a) Chest pain
- b) Pain in back, neck, jaw, arm, shoulder or back
- c) Dyspnea
- d) Cyanosis of lips, mucous membranes, and nail beds
- e) Dizziness

2) Nurse Assistant duties and observations

- a) Remain with patient/resident
- b) Notify licensed nurse immediately if above signs & symptoms are noted
- c) Monitor vital signs



List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

b. Myocardial Infarction (MI, heart attack)

1) Definition

- a) Clot in coronary artery causes a blockage of blood supply to the heart
- b) Damaged area can no longer function and turns to scar tissue

2) Signs and Symptoms

- a) Sudden, severe pain in chest usually described as crushing
- b) Radiating pain to arm, jaw, neck or back
- c) May present as indigestion or heartburn
- d) Diaphoresis, cool, clammy skin
- e) Dizziness
- f) Pallor/cyanosis
- g) Dyspnea (short of breath)
- h) Weak, irregular pulse
- i) Low blood pressure
- j) Loss of consciousness
- k) Nausea/vomiting

List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

l) Restless, anxious, feeling of impending doom

m) Denial

3) Nurse Assistant duties and observations

a) Notify licensed nurse immediately if above signs and symptoms are noted

b) Remain with patient/resident; place patient/resident in comfortable position; loosen clothing

c) Encourage patient/resident to rest

d) Reassure patient/resident that help is coming

e) Monitor vital signs

f) Do not give the patient/resident liquids or food

g) Prepare to transfer to acute care facility

c. Congestive Heart Failure

1) Definition

a) Occurs when the heart does not pump well enough to meet the body's needs

b) Normal cardiac output cannot meet the needs for activities of daily living

c) Can be left sided or right sided or both

List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

2) Signs and Symptoms

- a) Hemoptysis (spitting up blood)
- b) Cough
- c) Dyspnea
- d) Orthopnea (difficulty breathing while lying down)
- e) Fatigue
- f) Decrease in ability to exercise and be active
- g) Confusion
- h) Increased pulse
- i) Possible irregular heartbeat
- j) Increased frequency of urination at night
- k) Cyanosis
- l) Edema in extremities
- m) Moist respirations
- n) High blood pressure

List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

3) Nurse Assistant duties and observations

- a) Low sodium diet
- b) Raise head of bed
- c) Fluid restriction (if ordered)
- d) Anti-embolic stockings
- e) Assist with ADLs
- f) Range of motion to improve muscle tone
- g) Assist with O2 therapy
- h) Measure intake and output
- i) Measure daily weight
- j) Bedside commode as indicated
- k) Check vital signs: monitor apical pulse for one full minute; report if it drops below 60
- l) Pace activities

d. Peripheral Vascular Disease (PVD)

1) Definition

- a) Condition in which the blood supply to the legs, feet, arms, or hands is decreased due to poor circulation

b) Can be arterial or venous

Module 13: Long Term Patient/Resident

2) Signs and symptoms

- a) Painful cramping in the hips, thighs or calves when walking
- b) Painful cramping in the legs that does not go away

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List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

d. Peripheral Vascular Disease (PVD)

1) Definition

- a) Condition in which the blood supply to the legs, feet, arms, or hands is decreased due to poor circulation
- b) Can be arterial or venous

2) Signs and symptoms

- a) Painful cramping in the hips, thighs or calves when walking
- b) Painful cramping in the legs that does not go away
- c) Cyanotic hands or feet (blue or gray tinged)
- d) Arms and/or legs feel cool or cold to the touch
- e) Edema in the hands and feet
- f) Ulcers on the legs or feet (that are slow to heal or don't heal)
- g) Gangrene

3) Nurse Assistant duties and observations

- a) Encourage patient/resident to follow special diet/fluid restriction
- b) Record accurately intake and output
- c) Monitor for increased edema

List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

- d) Monitor vital signs report any changes (especially pulse and blood pressure)
 - e) Report to nurse any complaints of pain or discomfort in the hands, legs or feet
 - f) Report any type of chest, back, jaw, or shoulder pain or discomfort
 - g) Apply anti-embolic stockings (if ordered)
- e. Anemia; a condition that results from a decrease in the quantity of quality of red blood cells
 - 1) Signs and symptoms
 - a) Lethargy
 - b) Pale or jaundiced
 - c) Dyspnea
 - d) Digestive problems
 - e) Rapid pulse and increased respiratory rate
 - f) Cold
 - g) Dizzy
 - 2) Nurse Assistant duties and observations
 - a) Provide nutritional meals; increase iron containing foods like red meat and green leafy vegetables
 - b) Notify licensed nurse if signs of bleeding or black stool

 List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

- c) Monitor vital signs
 - d) Observe for and notify if above signs and symptoms appear or worsen
- f. Hypertension, high blood pressure
 - 1) Signs and symptoms
 - a) Blood pressure elevated above 140 systolic and/or 90 diastolic
 - b) Nosebleed
 - c) Eye hemorrhage
 - d) Dizziness
 - e) Trembling
 - f) Red complexion
 - g) Headache
 - h) Blurred vision
 - 2) Nurse Assistant duties and observations
 - a) Monitor blood pressure
 - b) Assist with stress management techniques
 - c) Low sodium diet

List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

- d) Assist with weight loss efforts
- e) Encourage regular exercise
- f) Notify licensed nurse if above symptoms are noted
- g. Other related subjects (optional): pacemakers, angina, varicose veins, atherosclerosis, internal defibrillation, anemia, blood types

D. Musculoskeletal System

1. Anatomy & Physiology

- a. Bones – hard rigid structures made of connective tissue cells
- b. Joints – point at which two or more bones meet; allow body movement
- c. Muscle – allows movement of body parts, maintains posture, and produces body heat

2. Functions

- a. Gives shape and form, maintains posture, permits movement, protects internal organs, stores calcium and phosphorus, produces heat and produces some blood cells
- b. Forms framework that supports the body and allows for movement



List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

3. Common diseases and disorders

a. Arthritis; inflammation of the joints

- 1) Rheumatoid; joint tissue lining is affected
- 2) Osteoarthritis; cartilage covering ends of bones deteriorates and ends of bones rub together, causing pain and deformity
- 3) Gout; metabolic disease caused by deposits of crystals at joint due to increase in uric acid
- 4) Signs and Symptoms
 - a) Pain at joint
 - b) Deformity at joint
 - c) Swelling
 - d) Tenderness
 - e) Heat at inflamed site
 - f) Fatigue
- 5) Nurse Assistant duties and observations
 - a) Notify licensed nurse if patient/resident requests medication for discomfort
 - b) Balance rest and exercise



List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

- c) ROM when no pain/inflammation, rest joint if pain/inflammation
 - d) Prevent contractures
- b. Fractures; break in the continuity of the bone
 - 1) Signs and symptoms (vary with type of fracture)
 - a) Pain, swelling, bruising at fracture site
 - b) Exposed broken bone through skin (compound fracture)
 - c) Immobility of area affected
 - d) Deformity
 - 2) Nurse Assistant duties and observations
 - a) Keep area immobilized (cast, splint)
 - b) Report pain
 - c) Reduce edema (elevate area affected)
 - d) Maintain alignment
 - e) Cast care (no breaks, chips, check distal pulses, check color, movement, and sensation distal to cast)
 - f) Report signs of infection



List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

c. Osteoporosis; caused by lost bone mass and results in porous, spongy bones that are easily fractured. Most common in elderly females

1) Signs and symptoms

- a) Fracture from very little or no trauma (often in hip or low back)
- b) Curvature of spine
- c) Loss of height
- d) Progressive weakness

2) Nurse Assistant duties and observations

- a) Prevent further fractures
- b) Report pain
- c) Gentle ROM
- d) Apply splints or braces as ordered
- e) Encourage high calcium meals (milk, dairy products, and green leafy vegetables)
- f) Mechanical lift for transfers
- g) Gentle handling and positioning

List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

d. Fractured hip; fracture of the femur, most frequent cause is falling and osteoporosis

1) Signs and symptoms

- a) Cannot get up after a fall
- b) Affected side is shortened and externally rotated
- c) Severe pain in hip or knee
- d) Edema in hip, thigh and groin

2) Nurse Assistant duties and observations

a) Pre-surgery hip precautions

- Avoid moving patient/resident until instructed by licensed nurse
- Use sheet or back board to move

b) Post-surgery hip precautions

- Do not flex hip more than 90 degrees
- Do not cross affected leg over midline of body
- Do not internally rotate hip on affected side
- Do not do passive ROM on affected side
- No weight bearing for several weeks after surgery



List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

- Use fracture bedpan
- Maintain proper hip alignment; may use trochanter roll and abduction pillow

4. Other related subjects (optional): fibromyalgia, traction, Continuous Passive Motion (CPM) machine, back injuries (ruptured disk), extremity amputation

E. Endocrine System

1. Anatomy and physiology
 - a. Pituitary; master gland
 - b. Thyroid; controls metabolism
 - c. Pancreas; secretes insulin to promote glucose use by cells
 - d. Adrenals; secretes adrenalin (fight or flight hormone)
2. Function; network of glands that secrete hormones into the blood stream
3. Common diseases and disorders
 - a. Diabetes Mellitus; chronic disease that results from a deficiency of insulin or a resistance to the effects of insulin
 - b. Body is unable to properly process food, and convert food into energy. There are two types of diabetes:
 - 1) Type 1; Insulin dependent diabetes mellitus (IDDM)

List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

2) Type 2; Non-insulin dependent diabetes mellitus (NIDDM)

3) Signs and symptoms

- a) The 3 polys: polyuria (excessive urination), polydipsia (excessive thirst), and polyphagia (excessive hunger)
- b) Fatigue
- c) Skin infections that are slow to heal
- d) Itching
- e) Burning on urination
- f) Vision changes
- g) Hyperglycemia; caused by inadequate insulin for metabolic needs
 - Develops slowly (over 24 hour period)
 - Confusion, drowsiness, slow slippage into coma
 - Headache
 - Sweet fruity odor to breath
 - Deep breathing
 - Low blood pressure
 - Nausea or vomiting



List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

- Flushed, dry, hot skin
- Unconsciousness
- Sugar in urine
- High blood sugar on glucose monitoring

h) Hypoglycemia; occurs when blood glucose is below normal; most commonly from overdose of insulin (insulin shock or insulin reaction)

- Occurs rapidly
- Hunger, weakness, dizziness, shakiness
- Moist and clammy
- Rapid, shallow respiration
- Nervous and excited
- Rapid pulse
- Unconscious
- No sugar in the urine
- Low blood sugar glucose monitoring

List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

4) Nurse Assistant duties and observations

- a) Know signs & symptoms of hyperglycemia and hypoglycemia
- b) Notify licensed nurse immediately if diabetic symptoms appear
- c) Offer meals and snacks at regular intervals, report uneaten portions
- d) Be aware that illness, stress, and infection can make blood sugars increase
- e) Observe extremities for infection, trauma or wounds (especially feet)
- f) Notify licensed nurse if patient/resident vomits after meal
- g) Offer easily assimilated source of carbohydrates if signs of hypoglycemia occur (after notifying licensed nurse)
- h) Make sure right diet is given
- i) Urine testing for sugar and acetone (rarely done in LTC now)
- j) Special foot care
 - Wash daily, dry between toes
 - Inspect feet for signs of irritation
 - Toenails to be cut by licensed nurse or podiatrist only
 - Shoes and stockings to be worn at all time when out of bed (never go barefoot)
 - Check for anything that impairs circulation

List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

- Remember: diet, exercise and medication are the most important components of diabetic treatment

4. Related subjects (optional): hyper and hypothyroidism, Cushing's and Addison's disease, glucose monitoring

F. Nervous System

1. Anatomy and Physiology (see handouts)

a. Brain (part of CNS)

- 1) Center of conscious thought and voluntary action
- 2) Motor coordination
- 3) Centers of control for respiration, heart function, and body temperature

b. Spinal Cord (part of CNS)

- 1) Transmission of impulses to and from brain
- 2) Approximately 17 inches long
- 3) Protected by spinal column (vertebrae)

c. Peripheral nerves

- 1) Transmission of impulses to and from spinal cord
- 2) Connects the CNS with the various structures of the body



List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

d. Sensory organs

- 1) Eye
- 2) Nose
- 3) Tongue
- 4) Skin
- 5) Ears

2. Functions

- a. Controls, directs, and coordinates functions of the body
- b. There are two major divisions; the central nervous system (CNS) and the peripheral nervous system (PNS)

3. Common diseases and disorders

- a. Dementia; any disorder of the brain that causes deficits in thinking, memory and judgment
 - i. Causes
 1. Alzheimer's disease (most common)
 2. Multi-infarction dementia (multiple TIAs)
 3. Parkinson's disease
 4. Syphilis

 List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

5. AIDS

6. Nutrition problems

7. Medication problems

8. Depression

9. Metabolic disorders

decreases ii. Alzheimer's disease (previously covered): caused by plaques and tangles in the brain's nervous system. Brain in size as cells are lost. Progressive and incurable

iii. Signs and Symptoms

Stage 1: Mild Dementia

- a. Short-term memory loss
- b. Personality changes
- c. Disorientation to time
- d. Poor judgment
- e. Lack of safety awareness
- f. Careless in appearance
- g. Anxious, depressed

 **List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.**

h. Delusions of persecution

Stage 2: Moderate Dementia

- Increased short-term memory loss
- Complete disorientation
- Wandering and pacing
- Sundowning
- Perseveration phenomena
- Motor skills deteriorate
- Speech, reading, writing deteriorates
- Incontinent of bowel and bladder
- Catastrophic episode

Stage 3: Severe Dementia

- Totally dependent
- Verbally unresponsive
- May have seizures



List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

- 4) Nurse Assistant duties and observations for patient/resident with dementia (all types)
 - a) Protect patient/resident from injury
 - b) Be compassionate, patient, and calm; maintain sense of humor
 - c) Encourage independence as long as possible
 - d) Provide mental and physical activities within patient's/resident's capabilities
 - e) Support patient's/resident's dignity and self-esteem
 - f) Provide structured quiet environment that is uncluttered
 - g) Use appropriate body language; patient's/resident's will "read" the staff's behavior and reflect the mood of the staff in their own behavior
 - h) Give one, short, simple direction at a time
 - i) Observe for signs and symptoms of physical ailments; patient/resident may be unaware of illness
 - j) Assist patients/residents to maintain a dignified, attractive appearance by helping with grooming and dressing



List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

- h) Monitor food and fluid intake
 - a. Do not use plastic feeding utensils
 - b. Provide nutritious finger foods when the patient/resident is unable to use utensils
 - c. Be aware of swallowing difficulties
 - d. Prepare food (butter bread, cut into bites) and encourage patient/resident to feed self as long as possible
 - e. Check food temperatures
 - f. Check patient's/resident's mouth after eating for "pocketing" (hoarding food in the cheeks)
 - g. Weigh patients/residents to detect patterns of weight gain or loss
- i) Use touch appropriately; surprising patient/resident can result in a catastrophic reaction
- j) Avoid using logic, reasoning or lengthy explanations
- k) When ability to speak is lost, patient/resident may communicate through non-verbal means
 - a. Biting, scratching, and kicking to express displeasure
 - b. Facial expressions and body language may express moods
 - c. Observe for what triggers agitation or anger and work on prevention
- l) Use distraction and diversion when patient/resident is agitated, take hand and calmly walk to another activity



List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

- m) Realize that people with dementia are not responsible for what they say
 - They cannot change
 - They are not aware of what they are doing
 - They have poor impulse control
 - Allow them to “save face” by preserving their dignity
 - No one really knows what is happening in the mind of people with dementia
- q) Be aware of wandering and pacing concerns
 - Keep record to detect patterns of wandering and pacing
 - Some triggers are as follows
 - Noise; keep quiet, avoid crowds
 - Boredom; give activities
 - Unmet physical needs; evaluate needs
 - Stress; create calm environment
 - Pain; evaluate for sources of discomfort
 - Hunger; offer frequent snacks
 - Thirst; encourage fluids



List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

- Need to use restroom; take to restroom at least every two hours
- Looking for companionship, security, or loved one; therapeutic touch, talk, consistency
- Physical restraints; feel threatened
- Looking for a state of mind, not physical location
- If they are in an area they shouldn't be, calmly take their hand and redirect them to a safe area
- Avoid large numbers of staff approaching patient/resident
- Use positive directions "stay inside" instead of "don't go outside"
- Use gentle persuasion
- r) Compliment positive behavior and accomplishments
- s) Encourage rocking chairs to diffuse nervous energy
- t) Be aware that "sundowning" (increased agitation and confusion in late afternoon or evening) can occur and how to prevent it
 - Avoid over-fatigue
 - Encourage patient/resident to remain awake as much as possible during the day
 - Evening meal should be at least two hours before bedtime
 - Eliminate caffeine



List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

- Evening activities should be quiet and calm
- Soft music is calming
- Give massage/back rub
- Light bedtime snack
- Follow bedtime routine (check with family members)
- Check lighting; shadows and reflections can be disturbing
- u) Discourage “pillaging and hoarding”
 - Label all patient’s/resident’s belongings to identify if they are taken by confused patient/resident
 - Check room daily for stale food
 - Keep patient’s/resident’s hands busy
 - Provide a “rummaging” drawer or box for patient/resident
- v) Use reality orientation (R.O.) to regain connection to environment
 - Especially effective in 1st stages of Alzheimer’s
 - Use clock and calendar at bedside
 - Call by name



List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

- Answer questions honestly, but do not give information they are unable to handle; for example
 - The patient/resident asks if their deceased husband is coming to visit
 - Nurse Assistant could respond by saying “tell me about your husband”
- Do not argue with patient’s/resident’s reality
- Use bulletin boards decorated for the season
- Keep curtains open during the day
- Make sure they have their glasses and hearing aids on
- Do not expect patient/resident to remember you - identify yourself, as needed

w) Use reminiscing therapy

- Encourage sharing memories of past events
- Use “prompting” questions to show interest in their history
- Use active listening skills
- Realize the need to reminisce increases as we age
- Reminiscing serves as a life review and may validate the worth of their life
- Acknowledge validity of feelings and emotions- be comfortable with patient’s/resident’s tears as well as laughter



List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

- Reminiscing helps people adapt to aging by maintaining self-esteem and working through personal loss
- x) Incorporate validation therapy into all your interactions with dementia patient/resident (maintains disorientated person's dignity by acknowledging the person's memories and feeling)
 - Developed by Naomi Feil
 - May be a way for patient's/resident's to work through issues from the past that were unresolved
 - Maintain identity and dignity of patient/resident by having pictures or reminders of who they once were in patient/resident room
 - Realize there is a reason for all behavior What appears to be confusion could be the person acting on an experience long ago
 - Disoriented people have a right to express emotions
 - Living must be resolved in order to prepare for dying
 - To live in reality is not the only way to live
 - Disoriented people have worth; they can still experience joy, pleasure and the ability to appreciate kindness
 - Within each confused person is a human being that was once a child and later an adult with hopes, joys, sadness, failures, and successes. They deserve to be respected, cared for, and loved in their final years

List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

- This therapy has been effective on lessening the need for restraints
 - y) Incorporate music therapy
 - z) Incorporate animal therapy
- b. Stroke (brain attack, cerebrovascular accident “CVA”): complete or partial loss of blood supply to cells of the brain
 - 1) Signs and Symptoms
 - a) Loss of sensation
 - b) Paralysis/hemiplegia
 - c) Aphasia
 - d) Dysphagia
 - e) Emotional lability
 - f) Loss of consciousness/confusion
 - g) Cognitive impairments
 - 2) Nurse Assistant duties and observations
 - a) Report changes in Level of Consciousness; headache or change in motor ability to licensed nurse
 - b) Sudden onset of symptoms indicates probable stroke
 - c) Prevent complications of immobility

List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

- d) Begin Restorative Care; assist with ADLs, encourage patient/resident to do as much as possible
- e) Be supportive emotionally. Patience is vital!
- f) Explore alternative methods of communication if unable to speak
- g) Be aware of swallowing/choking difficulties when feeding
- h) Incorporate reality orientation into care; clock, calendar, open curtains
- c. TIA (Transient Ischemic Attack) “mini stroke”
 - 1) Multi-infarction dementia (multiple Transient Ischemic Attack; TIAs)
 - 2) caused by lack of oxygen to the brain cells
 - 3) Signs & symptoms are temporary and subside when circulation is improved. (See CVA/Stroke) for possible signs & symptoms
- d. Parkinson’s Disease (refer to objective 4)
 - 1) Signs and symptoms
 - a) Tremors and “pill rolling”
 - b) Muscle rigidity
 - c) Difficulty and slowness in carrying out voluntary motor activities
 - d) Shuffling manner of walking

 **List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.**

- e) Slurred speech
- f) Loss of expressiveness in face
- g) Drooling
- h) Incontinence
- i) Constipation
- j) Urinary retention
- k) Mood swings
- l) Depression

2) Nurse Assistant duties and observations

- a) Maintain calm environment; symptoms will worsen when patient/resident is under stress
- b) Assist in ADLs
- c) Provide emotional support
- d) Restorative care; exercises
- e) Dementia care; as above



List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

e. Seizure disorder; recurrent, transient attacks of disturbed brain function (refer to objective 4)

1) Signs and symptoms

a) Generalized seizure (Grand mal)

- May experience “aura” or certain taste before seizure occurs
- Jerking and twitching of the body occurs
- Entire body is involved
- May experience loss of consciousness
- Incontinence
- May be followed by period of fatigue, confusion, disorientation

b) Partial seizure (Petit mal)

- Momentary loss of muscle tone
- May have periods of unconsciousness
- May chew, blink, or breathe rapidly
- Last 2-10 seconds
- “Absence seizures” in children can result in poor learning in school
- Only one extremity or part of an extremity may be involved



List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

c) Status epilepticus

- Seizure lasts for a long period
- Serious medical emergency; life threatening
- Death may result

2) Nurse Assistant duties and observations

a) During a seizure

- Stay with person and call for assistance
- Assist to lay down
- Do not restrain movements or put anything in mouth
- Move away objects that may injure a person
- Maintain airway by loosening clothing, turn head so saliva/emesis drains to one side, lift shoulder and allow head to tilt back
- Watch and observe so you can report type of seizure activity

b) After a seizure

- Allow to sleep
- Clean if incontinence has occurred

 List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

- Check vitals
- f. Vision impairments; following are common disorders of the eye that cause visual limitation and/or blindness
 - 1) Cataracts; the normally clear lens of the eye becomes cloudy
 - a) Signs and symptoms
 - Cloudy lens over the eye
 - Leading cause of vision loss in adults over the age of 55
 - b) Follow post-op protocol if surgery has been performed
 - 2) Glaucoma; increased pressure within the eye
 - a) Signs and symptoms
 - Eye pain
 - Difficulty adjusting to darkness
 - Unable to detect color
 - May see “halos”
 - Headache
 - Fatigue
 - Blurred vision



List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

3) Nurse Assistant duties and observations for visual impairments

- a) Announce yourself by name when entering the room
- b) Encourage television or radio listening
- c) Be extra careful in explaining what you are doing
- d) Describe the food you are going to feed them; is it hot? Cold? Describe food placement like the hands of a clock
- e) Do not disrupt environment; keep ADL supplies in same place and do not move furniture unless necessary
- f) Encourage independence

g. Hearing Impairments

- 1) Otosclerosis; progressive form of deafness, unknown cause
- 2) Signs and symptoms - progressive loss of hearing

4. Related subjects (optional): West Nile Virus, Autonomic Dysreflexia, Mad Cow Disease

5. Nervous system complications of inactivity

- a. Weakness, limited mobility
- b. Insomnia from sleeping during the day

List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

G. Gastrointestinal system

1. Anatomy and physiology

- a. Mouth, tongue, teeth
- b. Esophagus
- c. Stomach
- d. Small intestine
- e. Large intestine
- f. Liver and gall bladder
- g. Pancreas

2. Functions

- a. Extends from mouth to anus
- b. Responsible for breaking food into simpler substances that can then be used by the body cells for nutrition

3. Common diseases and disorders

- a. Malignancies; cancerous growths in GI tract that can cause obstruction, and may result in surgery and/or an ostomy
 - 1) Signs and symptoms
 - a) Vomiting

 List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

- b) Extreme abdominal discomfort
- c) Indigestion
- d) Constipation
- e) Blood in the stool
- f) Flatus
- g) No bowel movement at all

2) Nurse Assistant duties and observations

- a) Monitor bowel movements
- b) Report discomfort to licensed nurse
- c) Care of ostomy; artificial opening in the abdominal wall for elimination of solid waste and flatus • Standard precautions
 - Remove appliance gently to prevent irritation to skin (follow facility policy)
 - Empty bag and wash if reusable with soap and water
 - Secure clamp at base
 - Discard disposable bag in biohazard waste and replace with new bag
 - Observe stoma for redness, irritation, and skin breakdown (report to licensed nurse if this is noted)

List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

- Wipe area around stoma gently after appliance is removed and wash with mild soap and water
- Apply creams as ordered
- Fit opening of appliance to stoma as needed
- When applying appliance, seal well to prevent leaking. Observe color, character, amount and frequency of stools, and chart
- Be matter of fact, avoid looking repelled or disgusted (patient/resident may be self-conscious)

b. Common problems with elimination

1) Constipation

- a) Abdominal or rectal pain
- b) Inability to pass stool
- c) Loss of appetite
- d) Feel urge to defecate, but unable to pass stool
- e) Bloating
- f) Abdominal distention/hardness
- g) Liquid stool or mucus seeping from rectum
- h) Hard, dry stool

List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

- i) Feces that moves too slowly through the body
 - j) Caused by decreased fluids, diet, inactivity, ignoring the urge to defecate
- 2) Fecal impaction
 - a) Stool is unable to pass from the rectum
 - b) Results from unrelieved constipation
- 3) Diarrhea
 - a) Liquid or unformed stool with increased frequency
 - b) Feces that moves rapidly through the intestines
 - c) Caused by infection, medications, irritating foods
- 4) Bowel incontinence
 - a) Inability to control the passage of feces and gas
 - b) Possible cause due to injury or diseases of the nervous system or sphincter damage
 - c) May result when patients/residents do not receive the assistance they need in a timely manner
- 5) Flatulence (gas)
 - a) Excessive formation of gas in the stomach and intestines
 - b) Caused by foods, medications, or air swallowing

List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

c. Nurse Assistant duties and observations

- 1) Monitor and record bowel movements
- 2) Notify licensed nurse if above signs and symptoms develop
- 3) Encouraged high fiber diet for constipation
- 4) Encourage liquids whenever in contact with patient/resident
- 5) Encourage exercise to stimulate bowel activity
- 6) Prompt attention to call light
- 7) Monitor skin integrity
- 8) Prompt attention to incontinence care

a) NOTE: Nurse Assistants never perform removal of an impaction (digital disimpaction)

4. Related subjects (optional): Diarrhea, bowel incontinence, gall bladder disease, cholecystitis, hernias, ulcerations, bowel training, enemas, rectal tube, hemorrhoids, gastroesophageal reflux disease (GERD)

H. Urinary System

1. Anatomy and physiology

- a. Kidneys; filters the blood, forms urine



List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

- b. Ureter; tube carrying urine from the kidneys to the bladder
- c. Bladder; storage for urine
- d. Urethra; tube for passage of urine to outside during urination (voiding)

2. Functions

- a. Filters blood and produces urine in which excess fluids and toxins are excreted
- b. The body excretes 1000 to 1500 ml of urine/day
- c. Characteristics of normal urine
 - 1) Clear
 - 2) Amber (medium yellow color)
 - 3) Mild odor
 - 4) Approximately 1000 - 1500 cc in a 24-hour period

3. Common diseases and disorders

- a. Cystitis; inflammation of the urinary bladder, especially common in women
 - 1) Signs and symptoms
 - a) Dysuria, burning

 **List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.**

- b) Frequent urination
- c) Cloudy urine
- d) Hematuria
- e) Bladder spasm
- f) Loss of appetite
- g) Fever
- h) Confused patient/resident may exhibit agitation

2) Nurse Assistant duties and observations

- a) Encourage fluids
- b) Observe for normal urine
- c) Sitz bath
- d) Perineal care
- e) Always wipe front to back (from anterior to posterior)
- f) Rest

b. Urinary incontinence; loss of control of urine

List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

1) Signs and symptoms

- a) Unable to control urination
- b) Stress incontinence; urinate when exerting (sneezing, standing up, coughing)

2) Nurse Assistant duties and observations

- a) Bladder training
- b) Toilet patient/resident regularly
- c) Answer call light promptly
- d) Be positive when changing soiled linen or garments, do not be critical
- e) Perineal care
- f) Avoid liquids after dinner (if incontinent at night, or to prevent nocturia)

4. Related subjects (optional): renal dialysis, suppression, retention, renal calculi, specimen collection, urinary drainage systems, pyelonephritis, renal dialysis

 **List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.**

I. Reproductive System

1. Anatomy and physiology

a. Structures and organs (male):

- 1) Penis; tissue enlarges during sexual arousal to introduce sperm into vagina
- 2) Testes; secretes male hormone, produces sperm
- 3) Prostate gland; secretes fluid necessary for sperm activity
- 4) Epididymis; coiled tube on top of testis
- 5) Vas deferens; stores sperm

b. Structures and organs (female)

- 1) Ovaries; produce female hormone and eggs
- 2) Fallopian tubes; egg travels through to the uterus
- 3) Uterus; internal pear-shaped organ in the pelvis; holds fetus during pregnancy
- 4) Vagina; receives penis during sexual activity
- 5) Vulva; the external female genitalia

List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

2. Functions

- a. Reproduction of human life
- b. Production of some hormones

3. Common diseases and disorders

a. Male reproductive system

- 1) Enlarged prostate gland; urethra passes through prostate, and the prostate enlarges. It “strangles” urethra, causing difficulty in starting stream and in emptying bladder completely
- 2) Cancer of the prostate
- 3) Cancer of the testes; self-check of the scrotal sac should be done monthly

b. Female reproductive system

- 1) Cystocele; weakening of muscles between bladder and vagina
 - a) Signs and symptoms
 - Urinary incontinence
 - Frequent urinary tract infections
- 2) Rectocele; weakening of muscles between wall of rectum and vagina

List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

a) Signs and symptoms

- Hemorrhoids (varicose veins of the rectum)
- Constipation

b) Prolapsed uterus

3) Malignancies

- a) Breast
- b) Ovarian
- c) Uterine
- d) Cervical

c. Nurse Assistant duties and observations

- 1) Supportive care
- 2) Notify licensed nurse if distress occurs

4. Sexually transmitted disease

- a. Human Immunodeficiency Virus (HIV)/AIDS: viral infection transmitted primarily through direct contact with the bodily secretions of an infected person

List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

1) Signs and symptoms

- a) Flu-like symptoms
- b) Fever
- c) Night sweats
- d) Fatigue
- e) Swollen lymph nodes
- f) Sore throat
- g) GI upset
- h) Headache
- i) Kaposi's sarcoma
- j) Pneumocystis carinii pneumonia

2) Nurse Assistant duties and observations

- a) Be aware that virus is killed by 10:1 water to bleach solution; potency of solution good for three days
- b) Prevent secondary infections (immunosuppressed)
- c) Report new symptoms and complaints of discomfort to licensed nurse

List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

d) Provide comfort measures

5. Related subjects (optional): vulvovaginitis, tumors of reproductive tract, sexually transmitted diseases (STDs), breast self-exam, testicular self-exam

J. Immune System

1. Special cells and substances function to produce immunity
 - a. White blood cells
 - b. Antibodies
 - c. Antigens
 - d. "T" cells - type of cell that destroys invading cells
2. When the body senses an antigen, the immune system is activated
3. Functions to provide the body with immunity; protection against a disease or condition
4. Common diseases and disorders
 - a. Acquired Immunodeficiency Syndrome (AIDS); caused by a virus called Human Immunodeficiency Virus (HIV)
 - 1) Spread through certain body fluids
 - a) Blood



List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

- b) Semen
- c) Vaginal secretions
- d) Breast milk
- 2) Affects the body's ability to fight other viruses and bacteria
- 3) Signs and Symptoms
 - a) Loss of appetite
 - b) Weight loss
 - c) Fever
 - d) Night sweats
 - e) Diarrhea
 - f) Painful or difficulty swallowing
 - g) Tiredness, extreme or constant
 - h) Skin rashes
 - i) Swollen glands in the neck, underarms, and groin
 - j) Cough

 List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

- k) Sores or white patches in the mouth or on the tongue
- l) Purple blotches or bumps on the skin that look like bruises but do not disappear
- m) Confusion
- n) Forgetfulness
- o) Dementia
- 4) Nurse Assistant duties and observations
 - a) Practice standard precautions
 - b) Follow blood borne pathogen standard
 - c) Provide daily hygiene; avoid harsh soaps that irritate the skin
 - d) Provide oral hygiene before meals and at bedtime. Make sure the patient/resident uses a toothbrush with soft bristles
 - e) Provide oral fluids as ordered by licensed nurse
 - f) Measure and record intake and output
 - g) Measure weight daily
 - h) Have patient/resident perform deep breathing exercises as ordered
 - i) Practice measures to prevent pressure sores



List the body systems, including basic anatomy and physiology, common diseases of the elderly, signs and symptoms, and Nurse Assistant duties and observations.

- j) Assist with range-of-motion exercises and ambulation as ordered
- k) Encourage the patient/resident to perform self-care as able. The person may need assistive devices (walkers, commode, eating devices, etc.)
- l) Encourage the patient/resident to be as active as possible
- m) Change linens, gowns, or pajamas as often as needed when fever is present
- n) Be a good listener and provide emotional support

5) Other Considerations

- a) AIDS is often missed during diagnosis because aging can mask the signs and symptoms of AIDS
- b) Some persons infected with HIV don't develop AIDS for as long as 10-15 years and may not show signs/symptoms. However, they are carriers and can spread disease to others

Describe changes in body systems associated with aging

A. Integumentary System

1. Skin becomes thin, fragile, dry and wrinkled
2. Blood supply to fingers and toes decreases; sensitivity to hot and cold decreases
3. Hair loses color
4. Nails thicken
5. Bruises easily due to fragile blood vessels

B. Respiratory System

1. Lung strength decreases
2. Air sacs become less elastic and decrease in number
3. Airways become stiff and less elastic
4. Lung capacity decreases
5. Cough reflex is less effective
6. Cough becomes weaker
7. Gas exchange in lungs is less effective

Describe changes in body systems associated with aging

C. Cardiovascular System

1. Blood flow decreases leading to less efficient circulation
2. Blood vessels narrow and lose elasticity
3. Longer recovery to normal pulse after exercise
4. Heart is a less effective pump

D. Musculoskeletal

1. Muscles weaken and lose tone
2. Loss of elasticity in muscles
3. Loss of muscle mass causes weight loss
4. Less flexible
5. Bones lose minerals, become porous and brittle (easily broken)
6. Height is gradually lost due to space between vertebrae shrinking; posture slumped
7. Joints are less flexible and stiffer, slows normal body movement
8. Degenerative changes in joints
9. Decreased reflexes: prone to injury and falls

Describe changes in body systems associated with aging

E. Endocrine System

1. Decrease in sex hormones (estrogen and progesterone) signals menopause; testosterone levels decrease
2. Insulin production decreases
3. Metabolism and body functions slow
4. Body is less able to handle stress
5. Body less able to handle sugar
6. Metabolism and body functions slow

- F. Nervous System**
1. Slowing nerve transmission results in delayed reaction time, poor coordination and balance
 2. Less effective temperature regulation
 3. Decreased sensitivity to pressure and temperature changes
 4. Decreased blood flow to brain resulting in memory loss and confusion
 5. Deep sleep is shortened, naps are needed

Describe changes in body systems associated with aging

G. Gastrointestinal System

1. Decreased saliva production causing dysphagia
2. Taste buds not as sensitive
3. Decreased gag reflex leading to chance of choking
4. Slowed peristalsis leading to indigestion and constipation
5. Increased flatulence
6. Decreased digestive enzymes, avoid dry, fried and fatty foods
7. Decreased ability to absorb nutrients
8. Medication related to side effects
9. Sensitivity to sensation of full bowel
10. Urgency is exaggerated with the bowel

Describe changes in body systems associated with aging

H. Urinary System

1. Kidneys decrease in size and are less efficient
2. Bladder loses tone, leading to retention and infection (have patient/resident bear down after voiding to completely empty bladder)
3. Decreased bladder capacity leads to frequency of urination
4. Kidney function increases at rest, causing urination at night
5. Prostate enlargement in male causes frequency, dribbling, and urinary retention
6. Elderly patient/resident may exhibit behavior changes with usual UTI signs and symptoms
7. Decreased sensitivity to sensation of a full bladder
8. Urgency is exaggerated with the bladder

Describe changes in body systems associated with aging

I. Reproductive System

1. Male reproductive system

- a. Scrotum less firm
- b. Number and capability of sperm decreases
- c. Sexual response delayed
- d. Increase in size of prostate gland
- e. Hormone production decreases

2. Female reproductive system

- a. Menopause when menstruation ends
- b. Decrease production of estrogen and progesterone results in loss of calcium, causing brittle bones and osteoporosis
- c. Fewer hormones are produced
- d. Ovulation and menstrual cycle cease
- e. Vagina becomes thinner, drier
- f. Breast tissue decreases and muscles supporting breast weaken